
Practical - 4

Aim:

To Implement and Time analysis of factorial program using iterative and recursive method

Theory:

Both iterative and recursive implementations of the factorial program share a linear time complexity of $\mathcal{O}(n)$, but they differ drastically in their memory profiles. While the iterative approach uses constant space $\mathcal{O}(1)$, the recursive approach consumes linear space $\mathcal{O}(n)$ due to function call overhead

Algorithm

Algorithm (Iterative Method)

Start the program and read the value of n .

Initialize a variable $fact = 1$.

Repeat from 1 to n , multiplying $fact$ by each number.

Display the factorial value.

Stop the program.

Algorithm (Recursive Method)

Start the program and read the value of n .

If n is 0 or 1, return 1.

Otherwise, return $n \times \text{factorial}(n - 1)$.

Display the factorial value returned by the recursive function.

Stop the program.

Program Code

```
#include <iostream>
#include <chrono>

// Function for Iterative Factorial
// Time Complexity: O(n)
// Space Complexity: O(1)
unsigned long long factorialIterative(int n) {
    unsigned long long result = 1;
    for (int i = 1; i <= n; ++i) {
        result *= i;
    }
    return result;
}

// Function for Recursive Factorial
// Time Complexity: O(n)
// Space Complexity: O(n) due to call stack
unsigned long long factorialRecursive(int n) {
    if (n <= 1) return 1;
    return n * factorialRecursive(n - 1);
}

int main() {
    int n;
    std::cout << "Enter a non-negative integer (e.g., 20): ";
    if (!(std::cin >> n) || n < 0) {
        std::cerr << "Invalid input! Please enter a non-negative integer." << std::endl;
        return 1;
    }

    // Measure Iterative Implementation
    auto startIter = std::chrono::high_resolution_clock::now();
    unsigned long long resIter = factorialIterative(n);
    auto endIter = std::chrono::high_resolution_clock::now();
```

```
std::chrono::duration<double, std::nano> durationIter = endIter - startIter;

// Measure Recursive Implementation
auto startRec = std::chrono::high_resolution_clock::now();
unsigned long long resRec = factorialRecursive(n);
auto endRec = std::chrono::high_resolution_clock::now();

std::chrono::duration<double, std::nano> durationRec = endRec - startRec;

// Output Results
std::cout << "\n--- Results for " << n << "! ---" << std::endl;
std::cout << "Iterative Result : " << resIter << std::endl;
std::cout << "Iterative Time : " << durationIter.count() << " ns" << std::endl;
std::cout << "-----" << std::endl;
std::cout << "Recursive Result : " << resRec << std::endl;
std::cout << "Recursive Time : " << durationRec.count() << " ns" << std::endl;

std::cout<<"Shaik Abdul raheman
\n";
std::cout<<"92460118174\n";
std::cout<<"5-EN18\n";

return 0;
}
```

Result :

```
C:\Users\abdu\OneDrive\Doc × + ▾
Enter a non-negative integer (e.g., 20): 15

--- Results for 15! ---
Iterative Result : 1307674368000
Iterative Time   : 800 ns
-----
Recursive Result : 1307674368000
Recursive Time   : 200 ns
Shaik Abdul raheman
9246011874
5-En18

Process returned 0 (0x0)   execution time : 9.584 s
Press any key to continue.
```